Eng Math - II (Complex Analysis)

Practice Problem Set - 3

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1. Evaluate $\int_C \overline{z}\, dz$, where C is the right-half $z = 3e^{i\theta}$ $(-\pi/2 \le \theta \le \pi/2)$ of the circle $|z| = 3$ from $z = -3i$ to $z = 3i$.

2. Find an upper bound for $\left| \int_C e^{1/z}\, dz \right|$, where the integral is taken along the contour C, which is the quarter circle $|z| = 1,\ 0 \le \arg z \le \pi/2$ from the point 1 to the point i.

3. Let $z^{1/2}$ denote the branch $z^{1/2} = \sqrt{r}e^{i\theta/2}$ $(r > 0, -\pi/2 < \theta < 3\pi/2)$. Without actually finding the value of the integral, show that
$$\lim_{R\to\infty} \int_{C_R} \frac{z^{1/2}}{z^2+1}\, dz = 0,$$
where C_R denotes the semicircular path $z = Re^{i\theta}$ $(0 \le \theta \le \pi)$.

4. Integrate z^2 along (a) the line segment from 0 to i, (b) the arc of the parabola $y = x^2$ from 0 to $2 + 4i$, both directly and by finding a primitive of the integrand.

5. Evaluate $\int_C z^{1/2}\, dz$, where C denotes the semicircular path $z = 2e^{i\theta}$ $(0 \le \theta \le \pi)$ from the point $z = 2$ to the point $z = -2$.

6. Let $\sqrt{z}$ be the principal value of the square root of z. Evaluate $\int \frac{dz}{\sqrt{z}}$ along (a) the upper semicircle $|z| = 1$, (b) the lower semicircle $|z| = 1$. Why do they differ ?

7. Evaluate $\int_C |z|\, \overline{z}\, dz$ where C is the closed contour consisting of the line segment from -1 to 1 and the semicircle $|z| = 1,\ y \ge 0$, taken in the counterclockwise direction.

8. Assume $f(z)$ is holomorphic ($f'(z)$ continuous) and satisfies the inequality $|f(z) - 1| < 1$ throughout a domain D. Prove that
$$\int_C \frac{f'(z)}{f(z)}\, dz = 0$$
for every closed curve C in D.

9. For each of the following functions, examine whether Cauchy's theorem can be applied to evaluate the integrals around the unit circle taken counterclockwise. Hence or otherwise evaluate the integrals:

   (a) $\ln(z + 2)$,

   (b) $\dfrac{1}{|z|^3}$,

(c) $|z|$,

(d) e^{-z^2},

(e) $\overline{z}$,

(f) $\dfrac{1}{z^3}$.

10. Find:

(a) $\displaystyle\int_C \frac{z^2 - z + 2}{z^3 - 2z^2}\,dz$, where C is the boundary of the rectangle with vertices $3 \pm i$, $-1 \pm i$ traversed clockwise.

(b) $\displaystyle\int_C \frac{\sin z}{z + 3i}\,dz$, where $C : |z - 2 + 3i| = 1$ (counterclockwise).

11. Evaluate $\displaystyle\int_B f(z)\,dz$ where $f(z)$ is:

(a) $\dfrac{z+2}{\sin z}$,

(b) $\dfrac{z}{1 - e^z}$,

where B is the boundary of the domain between $|z| = 4$ and the square with sides along $x = \pm 1$, $y = \pm 1$, oriented in such a way that the domain always lies to its left.

12. C is the unit circle traversed counterclockwise. Integrate over C:

(a) $\dfrac{e^z - 1}{z}$,

(b) $\dfrac{z^3}{2z - i}$,

(c) $\dfrac{\cos z}{z - \pi}$,

(d) $\dfrac{\sin z}{2z}$.

13. Integrate $\dfrac{1}{z^4 - 1}$ over

(a) $|z + 1| = 1$,

(b) $|z - i| = 1$,

each curve being taken counterclockwise. *Hint: Resolve into partial fractions.*

14. Suppose that $f(z)$ is entire and its real part $u(x,\ y) = \text{Re}\,[f(z)]$ has an upper bound u_0 in xy- plane. Show that $u(x,\ y)$ must be constant throughout the plane.

15. Let f be continuous function on a closed bounded region R and let it be analytic and not constant throughout the interior of R. Assuming that $f(z) \neq 0$ anywhere in R, prove that $|f(z)|$ has a minimum value m in R which occurs on the boundary of R and never in the interior.

16. Consider the function $f(z) = (z+1)^2$ and the closed triangular region R with vertices at the points $z = 0,\ z = 2$ and $z = i$. Find points in R where $|f(z)|$ has its maximum and minimum values.

17. Find the Maclaurin series expansion of the function

$$f(z) \;=\; \frac{z}{z^2+9}.$$

18. Show that

$$\frac{1}{4z - z^2} \;=\; \frac{1}{4z} \;+\; \sum_{n=0}^{\infty} \frac{z^n}{4^{n+2}}, \qquad (0 < |z| < 4).$$

19. Find the Laurent series that represents the function

$$f(z) \;=\; z^2 \,\sin\left(\frac{1}{z^2}\right)$$

in the domain $0 < |z| < \infty$.

20. Find the singular points and isolated singular points of the function:

(a) $\frac{z+1}{z^3(z^2+1)}$

(b) $\text{Log}\, z \;=\; \ln r \;+\; i\,\theta \quad (r > 0,\ -\pi < \theta < \pi)$

(c) $\frac{1}{\sin\frac{\pi}{z}}$

21. Find the residue at $z = 0$ of the function

(a) $\frac{1}{z+z^2}$

(b) $z\, \cos\, \frac{1}{z}$

(c) $\frac{z - \sin z}{z}$

22. Use Cauchy's residue theorem to evaluate the integral of each of these functions around the circle $|z| = 3$ in the positive sense:

(a) $\frac{e^{-z}}{z^2}$

(b) $\frac{e^{-z}}{(z-1)^2}$

(c) $z^2\, e^{\frac{1}{z}}$

(d) $\frac{z+1}{z^2-2z}$